

THE GIT PROJECT REPORT

My GITHUB repository link :

<https://github.com/geee3/profile-project>

1) Add a Current Courses section to profile.html listing all the courses you are currently offering, then stage and commit with the message "Add list of current courses".

```
PS C:\Users\USER\profile-project> git log --online
```

```
3634aa1 (HEAD -> main) Add list of current courses
```

```
ec2c07a Add style.css and link to index.html
```

```
53cc437 Rename profile.html to index.html
```

```
da86170 Initial commit: create profile page
```

Screenshot below **Q3-5**

This screenshot shows the initial state of a project in VS Code. The Explorer sidebar on the left shows a folder named 'profile-project' containing a single file 'profile.html'. The main editor window displays the content of 'profile.html', which is an HTML document with a title 'My Profile' and several paragraphs of placeholder text. The bottom panel shows the Terminal with the following commands and output:

```
PS C:\Users\USER\profile-project> git add profile.html
PS C:\Users\USER\profile-project> git commit -m "Initial commit: create profile page"
[main (root-commit) da86170] Initial commit: create profile page
1 file changed, 15 insertions(+)
create mode 100644 profile.html
PS C:\Users\USER\profile-project> git log --oneline
da86170 (HEAD -> main) Initial commit: create profile page
PS C:\Users\USER\profile-project>
```

The right sidebar contains a 'Chat' panel with a 'Build with Agent' button and a 'Describe what to build' input field.

This screenshot shows the project after several updates. The Explorer sidebar now shows three files: 'index.html', 'style.css', and 'profile.html'. The main editor window displays the content of 'index.html', which includes the same placeholder text as 'profile.html' but also adds a section for 'Current Courses' with a list of four courses. The bottom panel shows the Terminal with the following commands and output:

```
PS C:\Users\USER\profile-project> git commit -m "Add list of current courses"
[main 3634aa1] Add list of current courses
1 file changed, 8 insertions(+)
PS C:\Users\USER\profile-project> git log --oneline
3634aa1 (HEAD -> main) Add list of current courses
ec2c07a Add style.css and link to index.html
53cc437 Rename profile.html to index.html
da86170 Initial commit: create profile page
PS C:\Users\USER\profile-project>
```

The right sidebar is now empty.

2) On the design update, merging it into the Main. {Q6-9}.

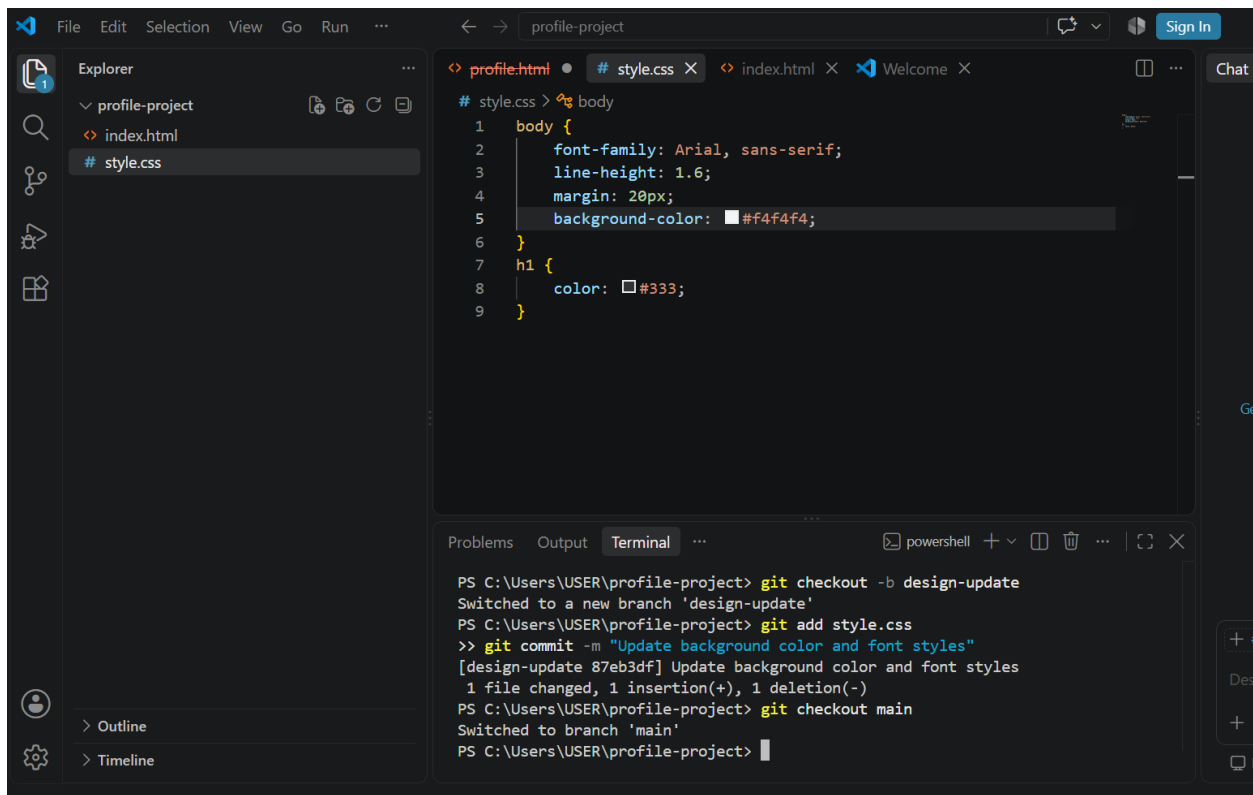
87eb3df (HEAD -> main, design-update) Update background color and font styles

3634aa1 Add list of current courses

ec2c07a Add style.css and link to index.html

53cc437 Rename profile.html to index.html

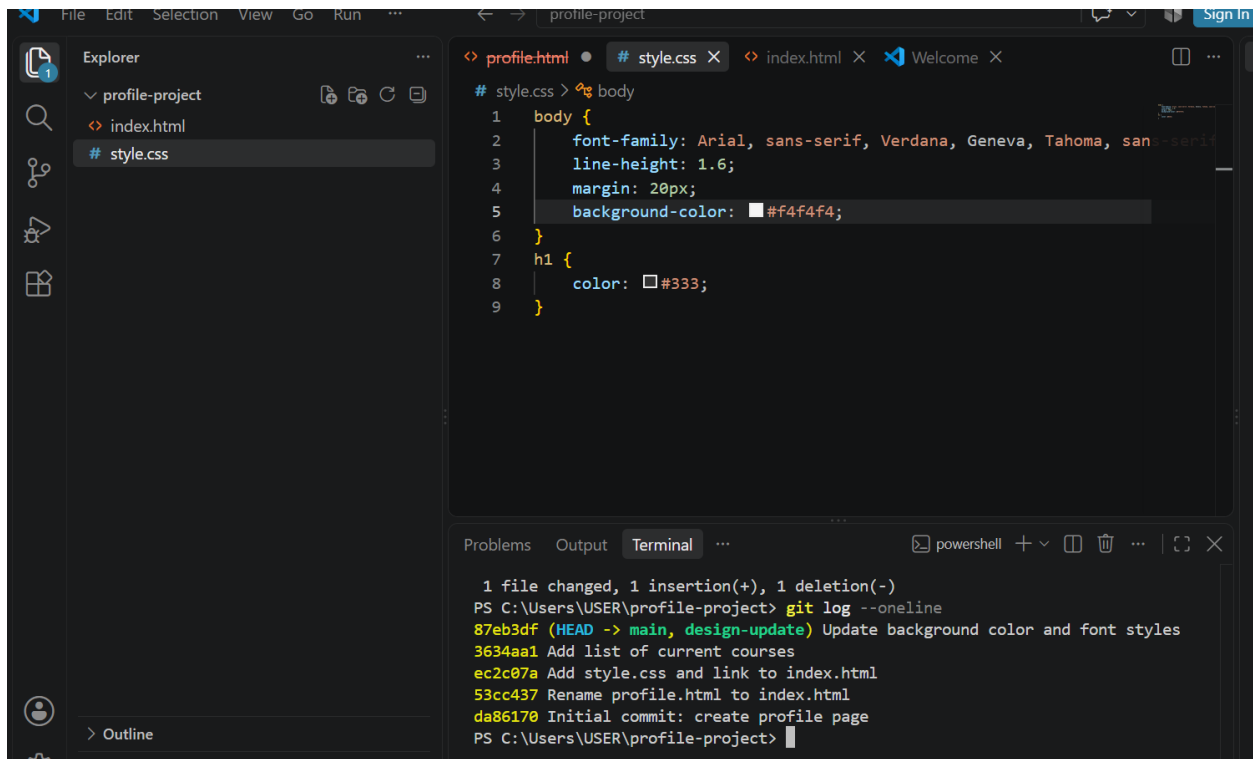
da86170 Initial commit: create profile page



```
File Edit Selection View Go Run ...
profile-project
index.html
style.css

# style.css > body
1 body {
2   font-family: Arial, sans-serif;
3   line-height: 1.6;
4   margin: 20px;
5   background-color: #f4f4f4;
6 }
7 h1 {
8   color: #333;
9 }
```

```
PS C:\Users\USER\profile-project> git checkout -b design-update
Switched to a new branch 'design-update'
PS C:\Users\USER\profile-project> git add style.css
>> git commit -m "Update background color and font styles"
[design-update 87eb3df] Update background color and font styles
1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\USER\profile-project> git checkout main
Switched to branch 'main'
PS C:\Users\USER\profile-project>
```



3)I confirmed that the cloned copy matches my local repository. Q10-11

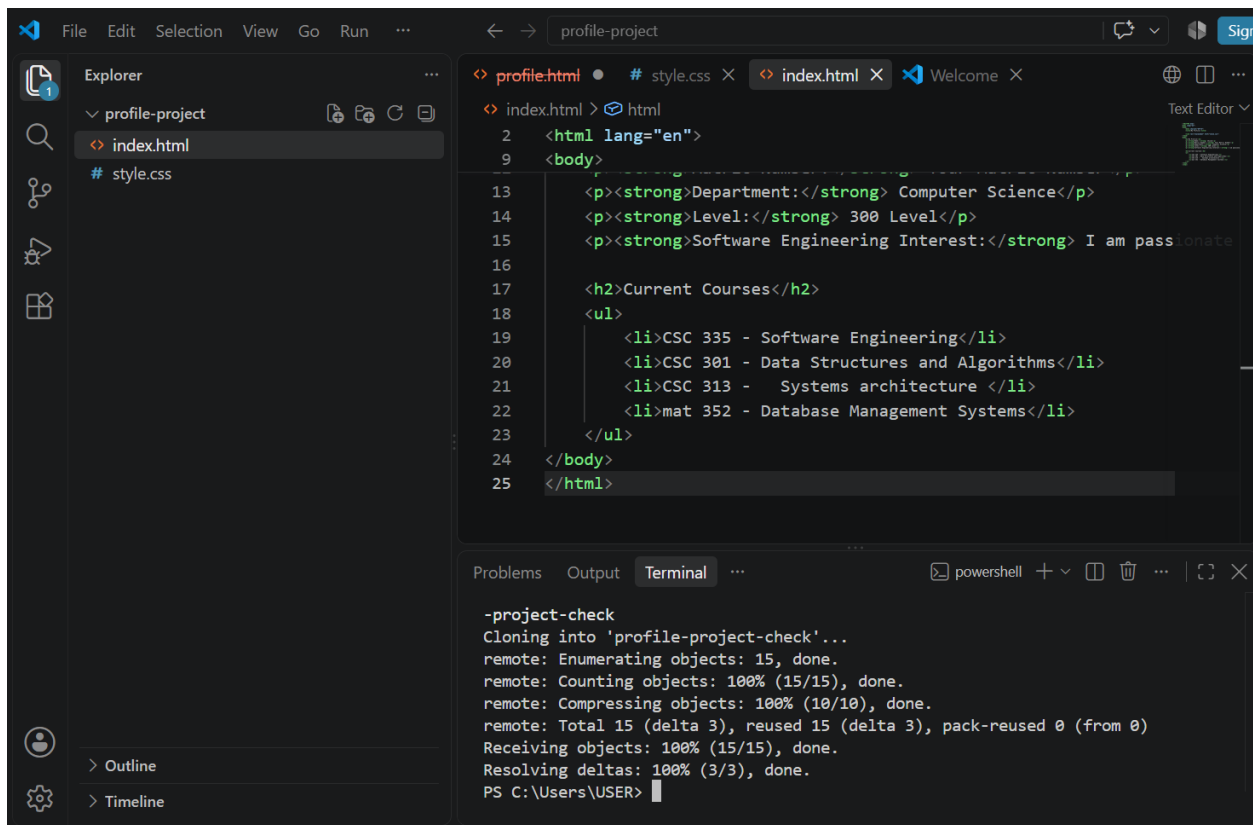
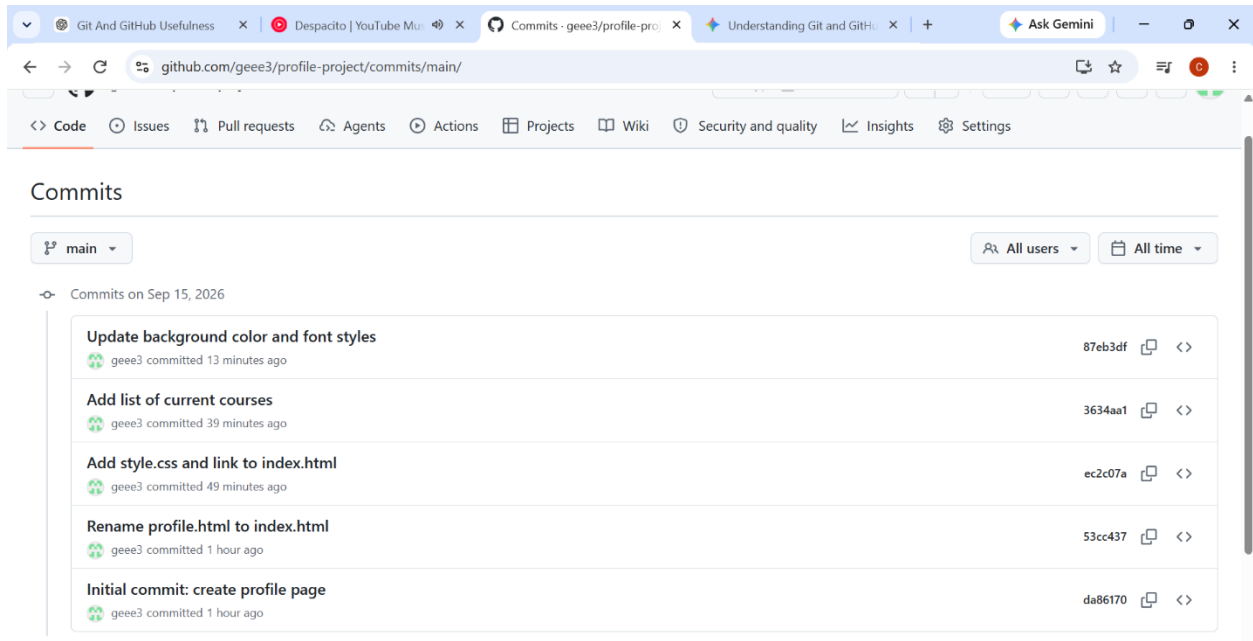
87eb3df (HEAD -> main, origin/main, origin/HEAD) Update background color and font styles

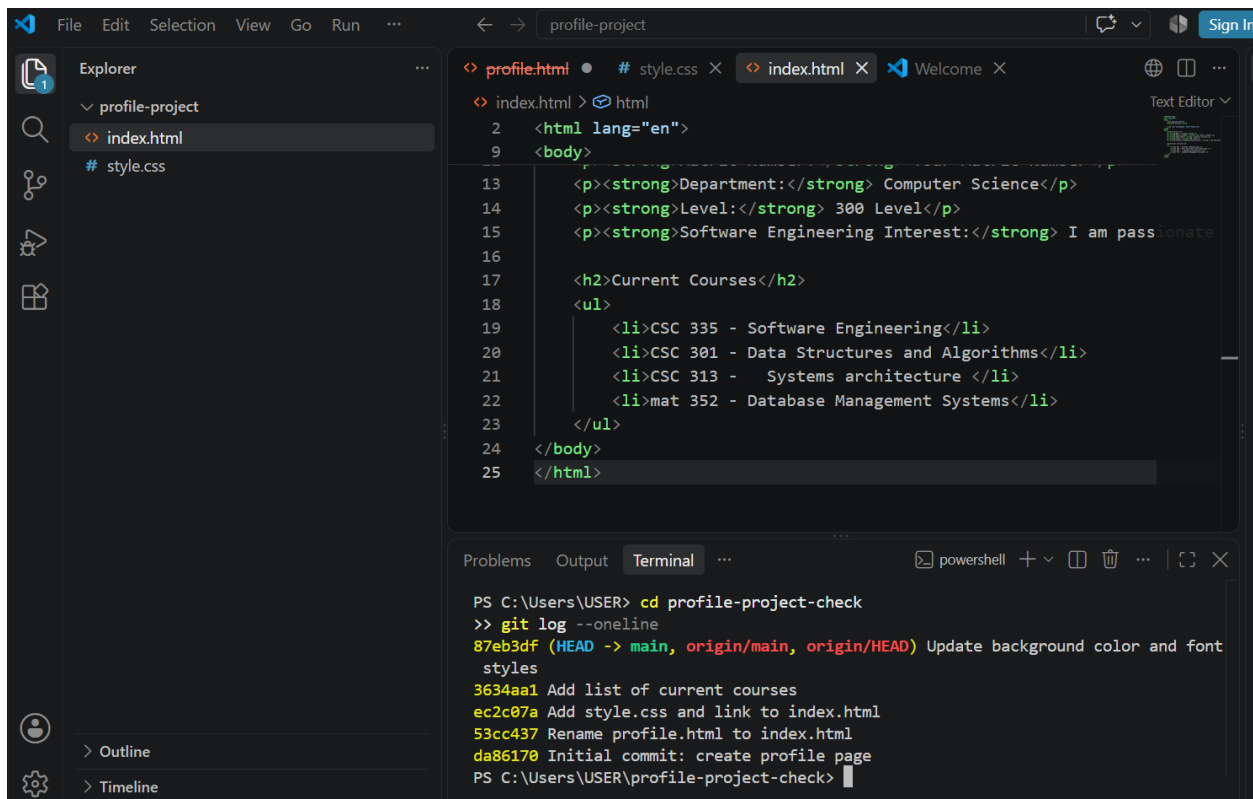
3634aa1 Add list of current courses

ec2c07a Add style.css and link to index.html

53cc437 Rename profile.html to index.html

da86170 Initial commit: create profile page





4) In your own words, briefly explain the difference between git add and git commit, and between git push and git pull. @Q12

git add: Imagine packing items into a box. It prepares your changed files for saving by putting them into a temporary holding spot (the "staging area"), but it hasn't sealed the box yet.

git commit: Imagine taping the box shut and sticking a label on it. It permanently saves a snapshot of everything you put in that box into your local project history.

git push vs. git pull

git push: Sending your saved box up to the cloud. It uploads your local commits from your computer to a remote server like GitHub.

git pull: Downloading a box from the cloud. It fetches the latest updates from GitHub and merges them straight onto your computer so your code stays up to date.

5) A new file called courses.html containing your current courses, course codes, and brief descriptions. Stage and commit the file.Q13

045fca9 (HEAD -> main, origin/main) Add courses.html with descriptions

87eb3df (design-update) Update background color and font styles

3634aa1 Add list of current courses

ec2c07a Add style.css and link to index.html

53cc437 Rename profile.html to index.html

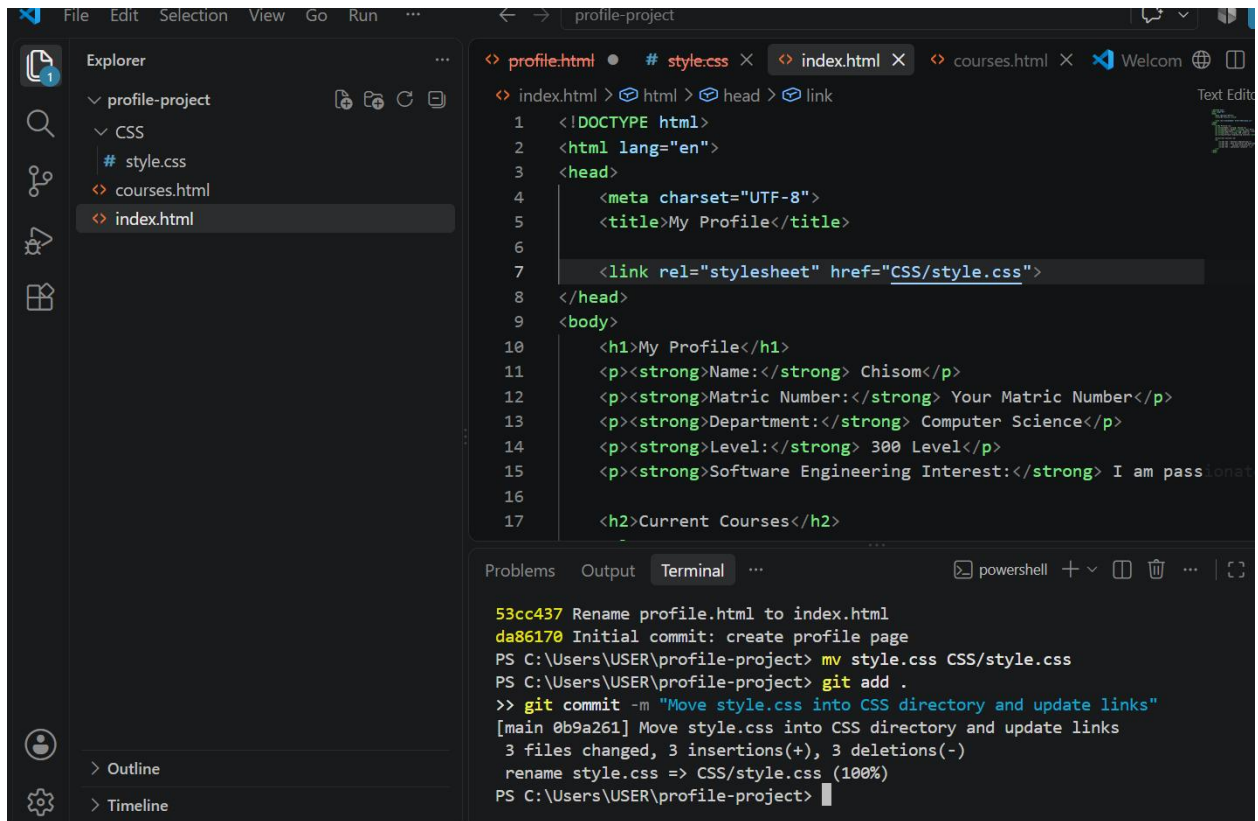
da86170 Initial commit: create profile page

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>My Courses</title>
6   <link rel="stylesheet" href="style.css">
7 </head>
8 <body>
9   <h1>Current Courses</h1>
10  <ul>
11    <li>
12      <strong>CSC 335 - Software Engineering:</strong>
13      Covers software development lifecycles, requirements anal
14    </li>
15    <li>
16      <strong>CSC 301 - Data Structures and Algorithms:</strong>
17      Focuses on data organization, abstract data types, search
```

Problems Output Terminal ... powershell + - [] [] ... []

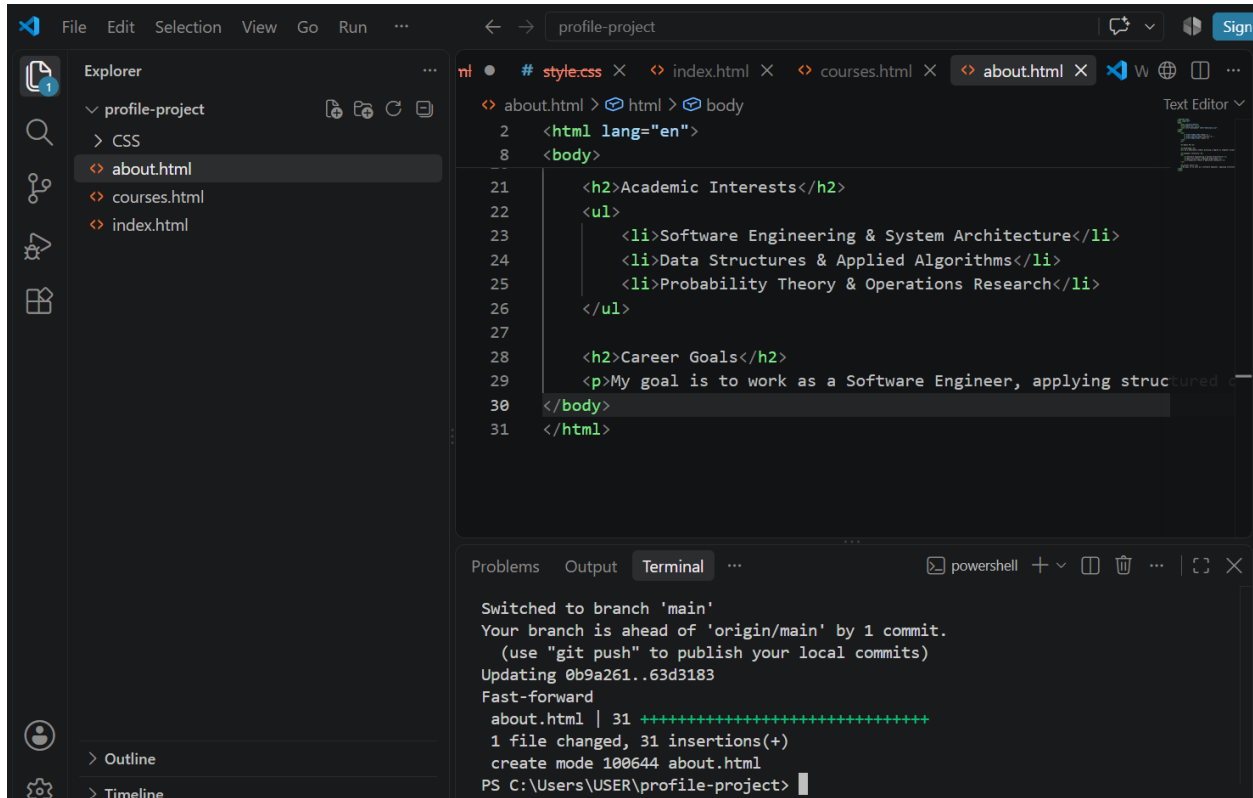
```
87eb3df..045fca9 main -> main
PS C:\Users\USER\profile-project> git log --oneline
045fca9 (HEAD -> main, origin/main) Add courses.html with descriptions
87eb3df (design-update) Update background color and font styles
3634aa1 Add list of current courses
ec2c07a Add style.css and link to index.html
53cc437 Rename profile.html to index.html
da86170 Initial commit: create profile page
PS C:\Users\USER\profile-project>
```


Screenshot for Q14&15 also:

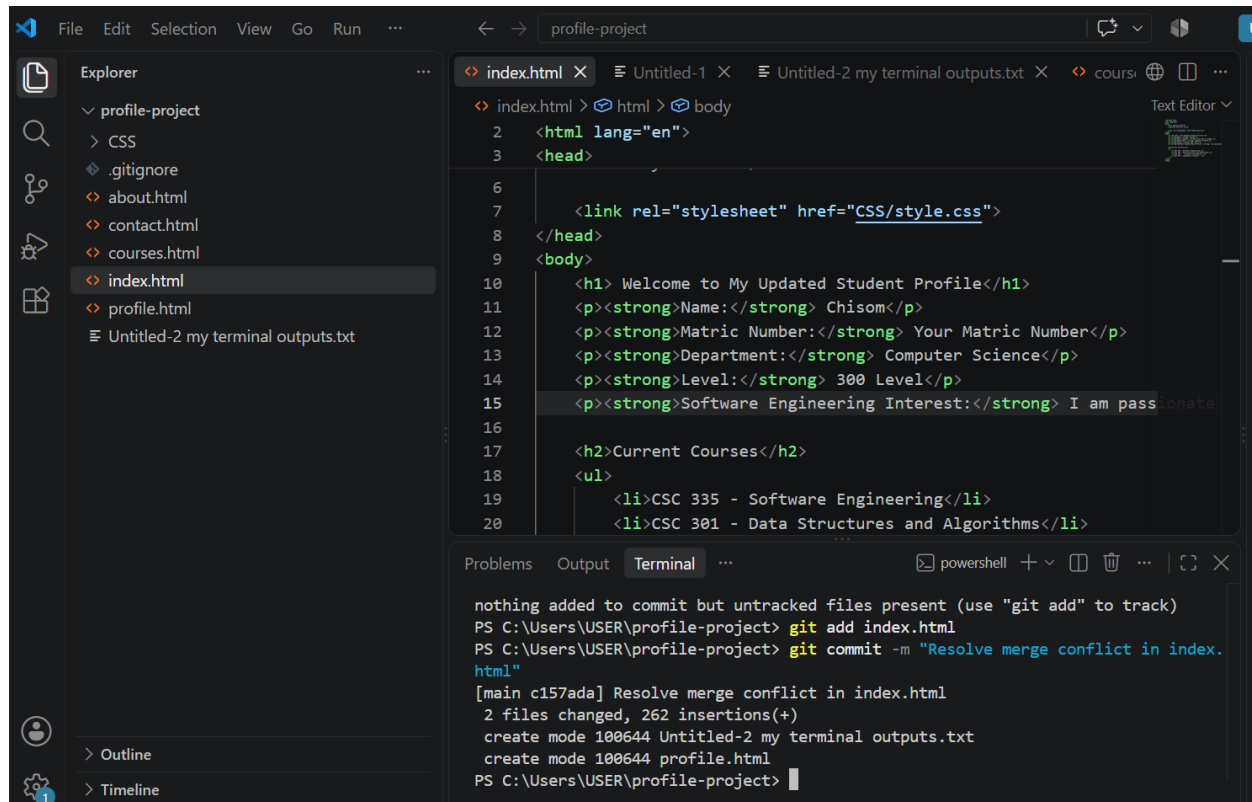


```
File Edit Selection View Go Run ...  
profile-project  
CSS  
# style.css  
courses.html  
index.html  
index.html > html > head > link  
1 <!DOCTYPE html>  
2 <html lang="en">  
3 <head>  
4 <meta charset="UTF-8">  
5 <title>My Profile</title>  
6  
7 <link rel="stylesheet" href="CSS/style.css">  
8 </head>  
9 <body>  
10 <h1>My Profile</h1>  
11 <p><strong>Name:</strong> Chisom</p>  
12 <p><strong>Matric Number:</strong> Your Matric Number</p>  
13 <p><strong>Department:</strong> Computer Science</p>  
14 <p><strong>Level:</strong> 300 Level</p>  
15 <p><strong>Software Engineering Interest:</strong> I am passionat  
16  
17 <h2>Current Courses</h2>  
Problems Output Terminal  
53cc437 Rename profile.html to index.html  
da86170 Initial commit: create profile page  
PS C:\Users\USER\profile-project> mv style.css CSS/style.css  
PS C:\Users\USER\profile-project> git add .  
>> git commit -m "Move style.css into CSS directory and update links"  
[main 0b9a261] Move style.css into CSS directory and update links  
3 files changed, 3 insertions(+), 3 deletions(-)  
rename style.css => CSS/style.css (100%)  
PS C:\Users\USER\profile-project>
```

6) Screenshot for Q16-18



7) Screenshot for 21&22



The screenshot shows the Visual Studio Code interface. On the left, the Explorer sidebar displays a file tree for a project named 'profile-project'. The files listed are 'CSS', '.gitignore', 'about.html', 'contact.html', 'courses.html', 'index.html' (selected), and 'profile.html'. Below the Explorer, the Outline and Timeline panels are visible. The main editor area shows the 'index.html' file with the following HTML code:

```
<html lang="en">
<head>
  <link rel="stylesheet" href="CSS/style.css">
</head>
<body>
  <h1> Welcome to My Updated Student Profile</h1>
  <p><strong>Name:</strong> Chisom</p>
  <p><strong>Matric Number:</strong> Your Matric Number</p>
  <p><strong>Department:</strong> Computer Science</p>
  <p><strong>Level:</strong> 300 Level</p>
  <p><strong>Software Engineering Interest:</strong> I am passionate
  <h2>Current Courses</h2>
  <ul>
    <li>CSC 335 - Software Engineering</li>
    <li>CSC 301 - Data Structures and Algorithms</li>
```

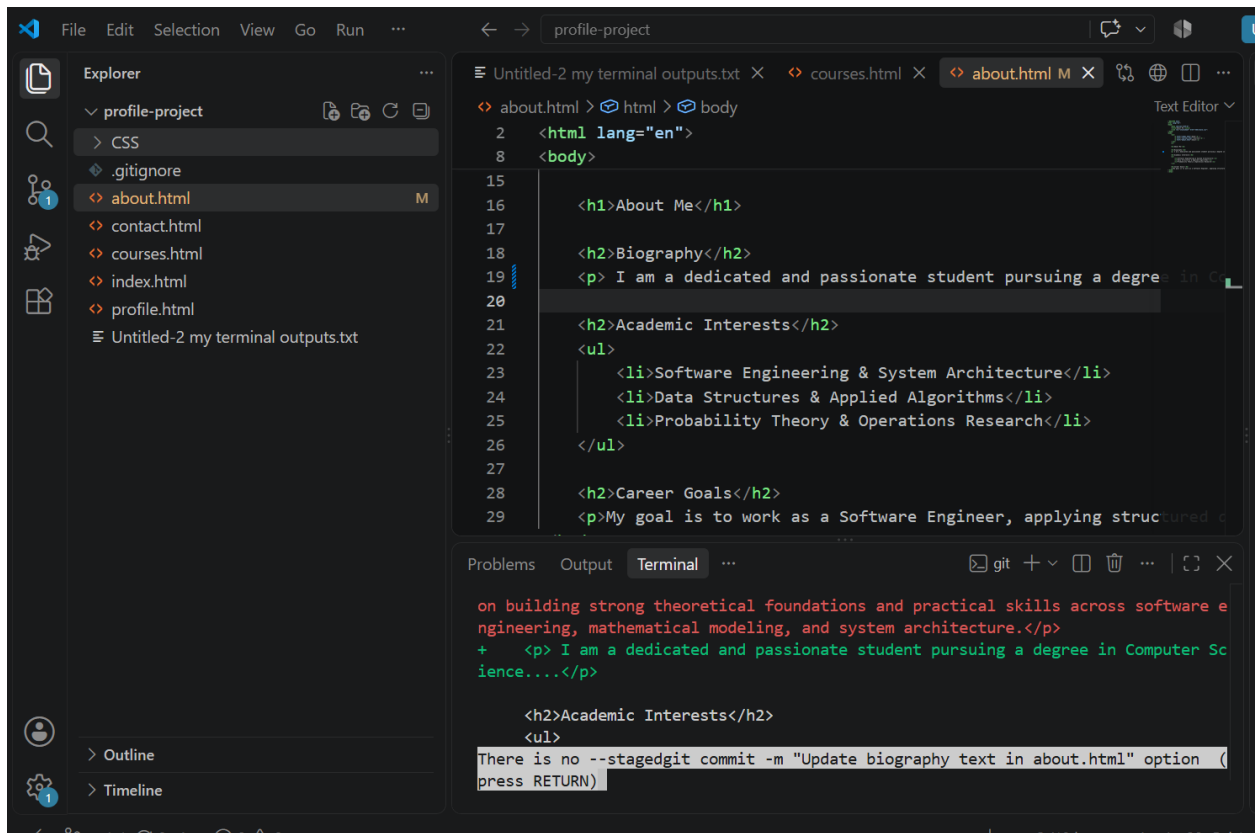
At the bottom, the Terminal panel shows the output of the following commands:

```
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\USER\profile-project> git add index.html
PS C:\Users\USER\profile-project> git commit -m "Resolve merge conflict in index.html"
[main c157ada] Resolve merge conflict in index.html
2 files changed, 262 insertions(+)
create mode 100644 Untitled-2 my terminal outputs.txt
create mode 100644 profile.html
PS C:\Users\USER\profile-project>
```

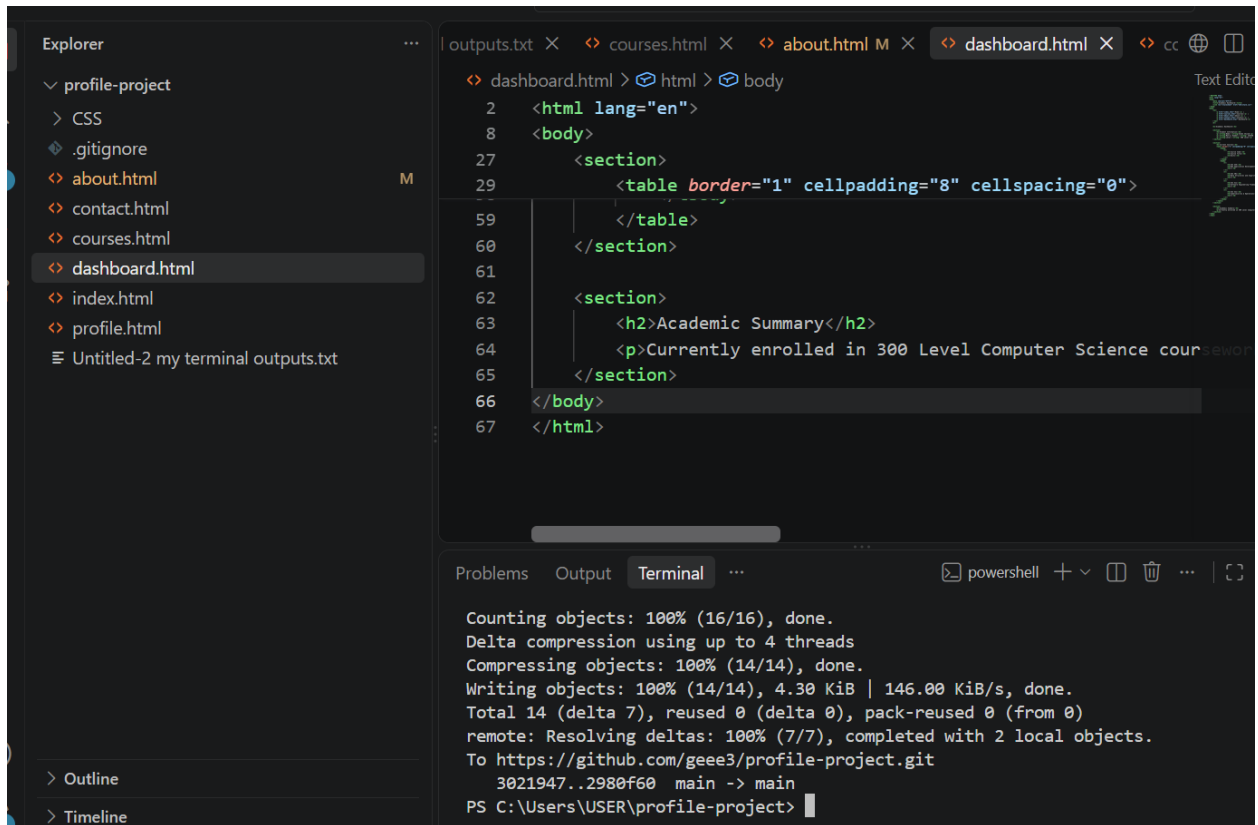
8) use git diff before staging it and git diff --staged after staging it. Explain the difference between the two outputs in your report.

You can include the following explanation in your assignment write-up:

- **git diff**: Compares the files in your **working directory** against the **staging area**. It shows modifications that have been saved to disk but have **not yet been staged** using git add.
- **git diff --staged** (or git diff --cached): Compares your **staged changes** against your **most recent commit (HEAD)**. It shows exactly what modifications are queued up and ready to be saved into the Git repository history upon running git commit.



9)Screenshot for Q24



10) Branch visual graph @25

- ```
* 2980f60 (HEAD -> main, origin/main, dashboard-feature) Add
dashboard.html with course table and academic summary

* c157ada Resolve merge conflict in index.html

* 00f61fb Merge branch 'profile-update'

|\
| * 11edc80 (profile-update) Update main heading on profile-update
branch

* | 21c361c Update main heading differently on main branch

|/
```

**Branch:** An isolated workspace in Git that allows you to build features or fix bugs without breaking your main project code.

**Merge Conflict:** A disagreement in Git that happens when two branches change the same line of code differently, requiring you to manually choose which code to keep.

**.gitignore:** A simple text file listing files or folders that Git should ignore, so clutter like temporary notes or build logs aren't saved into your project history.

**Local Repository:** The copy of your Git project and its commit history saved directly on your computer's hard drive.

**Remote Repository:** The online copy of your Git project hosted on a cloud platform like GitHub so you can back it up and share it with others.

